One Square Metre Observatory

SAVE NATURE



By

Tackling global environmental challenges, one square metre at a time.

SAVE NATURE, ONE SQUARE METRE AT A TIME

What if saving nature started with just one square metre?

The **One Square Metre Observatory** (m^2obs) invites everyone to become an urban nature researcher, turning everyday spaces into living laboratories for biodiversity and climate adaptation. From gardens and balconies to rooftops, schoolyards and public spaces, participants adopt one square metre to observe nature, document environmental changes and test simple ways to make their surroundings greener, cooler and more resilient.

Combining accessible observation methods, an easy-to-use digital platform and an open network of researchers and communities, **m^2obs** transforms individual observations into reliable, reusable knowledge. Scientific protocols, transparent data validation and human review help ensure that findings can support meaningful comparisons, while participants retain a voice in how their contributions are interpreted and shared.

Starting in The Hague and Budapest, we aim to build a citizen-powered movement that connects local observations with wider environmental knowledge, small experiments with practical solutions, and individual participation with collective action.

Your square metre tells the story of our planet.
Observe nature. Test small changes. Share what you discover. Together, save nature — one square metre at a time.

THE CHALLENGE & OPPORTUNITY

Climate change is global. Its impact is local. And urban areas are on the frontline.

Climate change is already transforming our cities. Rising temperatures, extreme rainfall, prolonged droughts and declining biodiversity are putting increasing pressure on urban nature and the quality of life of residents. Reducing emissions is essential. But mitigation alone is not enough. We must also adapt our cities to a changing climate, restore biodiversity and make our neighbourhoods greener, cooler and more resilient.
Yet we face two major challenges:

1. We lack sufficient local data.

Climate change affects every neighbourhood differently. But we still lack the detailed, long-term observations needed to understand what is happening at street level, how urban nature is responding and which local interventions can make a difference.

2. We need everyone to participate.

Governments and scientists cannot address these challenges alone. Lasting change requires the active involvement of residents, schools, communities and businesses. Yet many people lack accessible tools, knowledge and opportunities to contribute meaningfully.

The opportunity: turn citizens into urban nature researchers. By empowering people to observe, measure and improve just one square metre, we can build a collective understanding of our changing urban environment — while inspiring practical action to protect and restore nature.

THE ONE SQUARE METRE OBSERVATORY IN 4 STEPS



1 OBSERVE NATURE

Track temperature, rainfall, vegetation and biodiversity in their own square metre.



2 UNDERSTAND CHANGE

Discover how nature and local climate conditions differ between neighbourhoods, seasons and urban environments.



3 TAKE ACTION

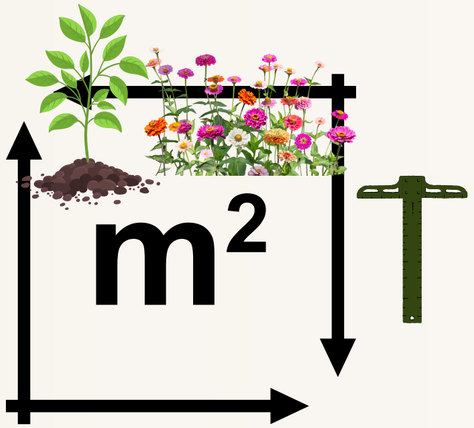
Explore simple, nature-friendly interventions and monitor how their square metre changes over time.



4 SHARE KNOWLEDGE

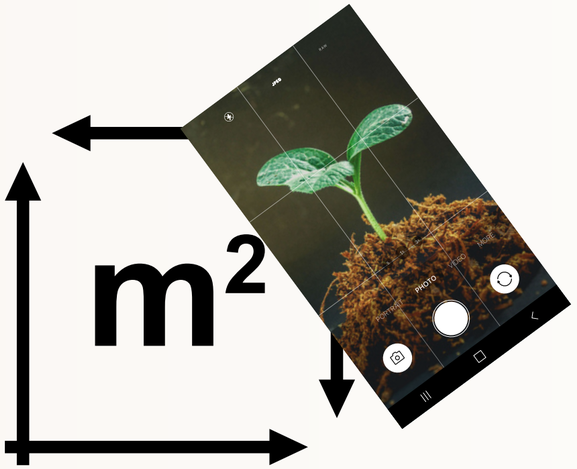
Contribute to an open, scientifically supported database that helps citizens, researchers and cities better understand and protect urban nature.

HOW IT WORKS



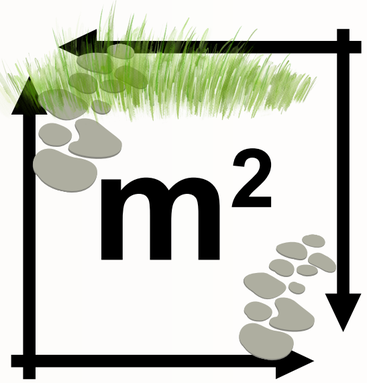
1 MARK YOUR SQUARE

Use four one-metre edges to define a standard observation plot. Mark the halfway points to create a simple grid.



2 OBSERVE AND PHOTOGRAPH

Record plants, flowers, pollinators, vegetation height, rainfall or soil conditions. Repeat photographs document change.



3 TEST AND COMPARE

Lay out the reusable kit on grass or soil. Compare a mown lawn with a meadow, or bare soil with a mulched patch.

SIMPLE TOOLS POWERFUL INSIGHTS

Save the Earth m² makes monitoring nature easy, affordable and accessible to everyone through a simple DIY measurement kit and an interactive online app.

THE m² MEASUREMENT KIT

A simple, low-cost kit made from four PVC profiles, a measuring tape, string and markers — all available at local hardware stores. Optional tools, such as a rain gauge, measuring rod or homemade infiltration ring, allow participants to explore nature in greater detail.

THE ONLINE APP

An easy-to-use digital platform guides participants step by step through their observations. Residents can register their square metre, upload photos, record measurements, receive reminders, track changes over time and discover how their results compare with others across the city.
The app also provides practical tips, nature-friendly challenges and insights into how small actions can help biodiversity thrive.

OPEN TO EVERYONE

All DIY building instructions and scientific observation protocols will be freely available online. Participants can build their own kit, use a ready-made kit or join measurements at public observation sites.
No expensive equipment. No scientific background required. Just one square metre, curiosity and the willingness to help nature thrive.

RESEARCH QUESTIONS PARTICIPANTS CAN EXPLORE

Participation empowers everyone to explore real environmental challenges, collect valuable data and discover how small changes can help urban nature thrive.

1. CAN WE HELP TREES SURVIVE DROUGHT?
Explore how trees retain water by monitoring soil conditions, measuring moisture and testing simple water-retention solutions around urban trees.

2. CAN WE CREATE MORE SPACE FOR POLLINATORS?
Count flowers, bees, butterflies and other pollinators. Discover how planting, mowing and natural vegetation influence biodiversity in gardens and public spaces.

3. CAN GREENERY MAKE OUR CITIES COOLER?
Compare temperatures across pavement, gravel, grass and planted areas. Explore how vegetation and nature-based solutions can help reduce urban heat.

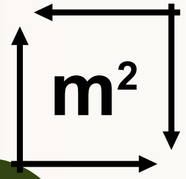
4. CAN WE HELP RAINWATER REACH THE SOIL?
Measure rainfall and test how quickly water infiltrates different surfaces. Discover how healthier soil and greener spaces can improve water absorption.

PARTICIPANT JOURNEY



1

Sign up and join the movement



2

Adopt your square metre



3

Set up your obsesrvatory



4

Observe, measure and upload your finding



5

Discover and learn



6

Take action and share your impact



7

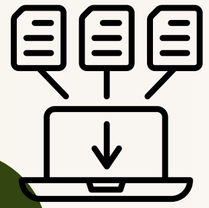
Receive feedback on data and tips to save nature around you



8

Invite others to participate

DATA JOURNEY



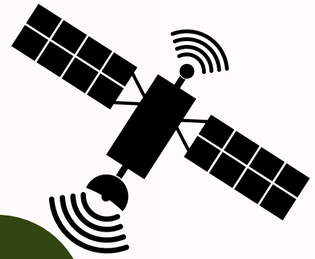
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Receive data from participants



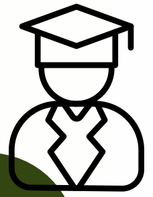
2

Collect & Secure



3

Validate and Verify



4

Analyse and understand



5

Visualise and Share



6

Turn knowledge into action



7

Measure progress

DATA JOURNEY AFTER RECEIVING DATA

COLLECT & SECURE DATA
Observations are stored securely, with personal information separated from measurement data. Each observation retains its location, time, method and source. Participants and participating organisations have clear rights over access, correction, publication and reuse.

VALIDATE & VERIFY
Automated checks identify missing, inconsistent or unusual measurements. Human reviewers assess data quality, compare observations with reference data where appropriate, and document corrections and uncertainties.

ANALYSE & UNDERSTAND
Researchers and communities explore patterns across locations, seasons and urban environments. Reproducible analyses help identify environmental changes, biodiversity trends and possible relationships between local interventions and observed outcomes.

9VISUALISE & SHARE
Reviewed observations and findings are presented through accessible maps, dashboards and reusable datasets. Publication respects privacy, agreed permissions and community governance, and clearly communicats data quality and scientific limitations.

TURN KNOWLEDGE INTO ACTION
Residents, nature organisations, researchers and municipalities use the evidence to identify local challenges, co-design nature-based interventions and make informed decisions. They remain in controll for interpreting findings and determining actions.

MEASURE PROGRESS
Repeated observations document environmental changes over time. Carefully designed comparisons help assess whether interventions are effective, distinguish observed improvements from uncertain causal effects, and guide further adaptation.

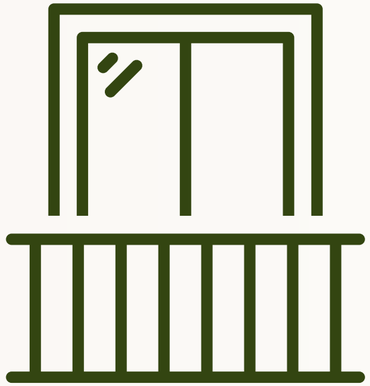
WHO WE ENGAGE (RESIDENTS)

Everyone can become an urban nature researcher.
Save the Earth m² makes citizen science accessible to everyone — whether you have a garden, a balcony or no outdoor space at all.



1 RESIDENTS WITH GARDEN

Adopt a square metre in their own garden to observe nature, measure local climate conditions and experiment with nature-friendly improvements.



2 BALCONY & ROOFTOP RESIDENTS

Turn even the smallest outdoor space into a nature observatory using specially designed measurement kits.



3 RESIDENTS WITHOUT PRIVATE OUTDOOR SPACE

Participate through designated public observation sites in parks, neighbourhoods and urban green spaces, individually or along nature discovery routes.

WHO WE ENGAGE (ORGANISATIONS)

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1 SCHOOLS and STUDENTS
Transform a piece of a schoolyards into outdoor classrooms where young people explore biodiversity, climate and scientific observation through hands-on learning.



2 COMMUNITY GROUPS & ORGANISATIONS
Mobilise neighbourhood associations, environmental volunteers and local businesses to adopt shared observation sites and organise collective nature activities.



3 NATURE & ENVIRONMENTAL ORGANISATIONS
Engage their members, volunteers and local networks in citizen science, support biodiversity monitoring, share ecological expertise and turn observations into collective nature conservation and restoration efforts.

OUR PARTNERSHIP ECOSYSTEM

PARTNER ROLE	PROPOSED PARTNERS	KEY CONTRIBUTIONS
TECHNOLOGY, PROGRAMME & COMMUNICATIONS	REPREX and hetkanWEL	Programme leadership, campaign, branding, App/PWA, API, hosting, data security
CITIES FOR PILOT	Municipalities of The Hague & Budapest	Public observation sites, environmental and spatial data, permits, links to urban planning and climate adaptation policies.
NATURE & EDUCATION	IVN Staatsbosbeheer	Environmental education, volunteers, training, observation activities, schools and permanent observatories.
SCIENCE	University or research institutes	Scientific observation protocols, measurement validation, research design, environmental analysis and impact evaluation.
SPONSORING	Local companies	Sponsorship of measurement kits, public observatories, educational activities and long-term community participation.
LOCAL COMMUNITIES	Schools, libraries, community centres	Participant recruitment, kit lending, local observations, community data stewardship, feedback and practical action.

REPREX

niAWARD-WINNING OPEN DATA INNOVATION
Winner of The Hague Innovators Audience Prize (2022), Reprex develops open-source, community-curated data platforms providing trustworthy information for research and AI.

OPEN-SOURCE SCIENTIFIC SOFTWARE
Reprex maintains software for reconciling financial, environmental and social data with authoritative sources, and for human review of AI-generated information. Its software undergoes independent review, is presented at scientific conferences and serves thousands of users.

DATA OBSERVATORIES
Through competitive tenders, Reprex developed European Music Observatory prototypes connecting public and private data across universities, libraries, rights organisations and the music industry. Other observatories cover sustainability and cultural heritage.

EUROPEAN KNOWLEDGE INFRASTRUCTURES
Reprex collaborates as a recognised data and software development partner with Wikimedia-related organisations, Europeana and the European Collaborative Cloud for Cultural Heritage, supporting interoperable, community-curated knowledge.

SEMANTIC DIGITAL TWINS
Reprex develops digital twins connecting the social and environmental histories of buildings, parks and urban spaces with their physical characteristics, making information inspectable for climate adaptation and resilience.

WIKI MUSEUM & COMMUNITY KNOWLEDGE
With Central and Northern European Wikimedia organisations, Reprex contributes to Wiki Museum, an open platform for sharing independently curated exhibitions across natural sciences, social sciences and the humanities.

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By

Joining? Please contact

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